

PROBLEMS IN DIFFERENTIATION

Limits, L'Hospital's Rule & Differentiability

PART 1

(a) $\lim_{x \rightarrow \infty} \frac{x - \sin x}{2x + \sin x}$

(b) $\lim_{x \rightarrow \infty} \frac{2x + \sin 2x + 1}{(2x + \sin 2x)(\sin x + 3)^2}$

(c) $\lim_{x \rightarrow 0^+} \left(2 \sin \sqrt{x} + \sqrt{x} \sin \frac{1}{x} \right)^x$

(d) $\lim_{x \rightarrow 0} \left(1 + x e^{-\frac{1}{x^2}} \sin \frac{1}{x^4} \right)^{e^{\frac{-1}{x^2}}}$

PART 2

Is the function given by

$$f(x) = \begin{cases} \frac{1}{x \ln 2} - \frac{1}{2^x - 1}, & x \neq 0, \\ \frac{1}{2}, & x = 0 \end{cases}$$

differentiable at zero?

Think • Apply • Solve • Master